

Managing and Monitoring Your DFSMSHsm Environment with Mainstar: HSM Reporter/Manager

Session #2437



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DFSMSHsm's Vital Role

- DFSMSHsm manages a large percentage of a data center's data
 - Installations report between 30-50% percent of their data is managed by DFSMSHsm migration
 - The majority of installations migrate data required to execute weekly, monthly, quarterly, and annual batch applications
 - More than 80% of installations migrate data they will need at the disaster recovery site in the event of a disaster

DFSMShsm's Vital Role

- Companies rely on data managed by DFSMShsm
 - Long-term data
 - Satisfy government requirements
 - Customer records, statistics, medical histories
 - Equipment drawings, plans, schematics
 - Reports
 - Purchasing histories, demographics
- Companies rely on data backed up by DFSMShsm
 - Often times, the only backup copy of a critical data set is managed by DFSMShsm

Native DFSMSHsm Reporting

- DFSMSHsm Activity Logs
 - Daily Space Management
 - Automatic data set migration
 - Multiple logs/day
 - Identify data sets failing migration
 - No reports, manual inspection
 - Daily Availability Management
 - Automatic data set backup
 - CDS backups
 - One log/day
 - Identify data sets failing backup
 - No reports, manual inspection

Native DFSMShsm Reporting

- DFSMShsm Activity Logs
 - Daily ABARS backups
 - Application Technology Recovery backups
 - Many logs/day
 - Identifies data sets backed up
 - Results collected by Mainstar Backup and Recovery for z/OS ABARS Manager feature
 - Reports available using Mainstar Backup and Recovery for z/OS ABARS Manager feature
- DFSMShsm Activity Failure Alerts
 - Operator console messages
 - Scheduler messages

Native DFSMShsm Maintenance

- DFSMShsm Activity Logs
 - Browse the logs to identify failures
 - Data set names and return codes
 - Determine cause of failure
 - Messages and Codes manual
 - Develop and execute corrective action
 - Batch jobs and/or commands
 - One data set at a time



DFSMSHsm Concerns

- Errors in DFSMSHsm Daily Operations
 - DFSMSHsm errors are not just in the CDS
 - High recall vs. migration thrashing
 - Volumes not touched by space management
 - Data sets not migrated
 - Data sets not backed up
 - Bad return codes from backup or migration
 - *Indicating data set errors*
 - Un-referenced data not migrating

DFSMSHsm Challenges

- Reporting on migrated data is difficult
 - Is data getting migrated
 - Does the data have the right DFSMS attributes
 - Who owns what data
- No means of determining thrashing
- Native DFSMSHsm audits take too long to perform
- Audits have not performed in years
 - As a result, there are likely many errors which represent data that can not be accessed
 - Which in turn, could represent a **single point of failure!**

Why You Need HSM Reporter/Manager



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HSM Reporter/Manager (HRM)

- Easily collect and display critical DFSMShsm information
 - Report on all DFSMShsm activity
 - Zoom-in on problem areas
 - Provides daily and historical information
 - Take corrective actions for problems detected before they become issues
 - Provides Health Reports that alert the administrator when DFSMShsm automatic tasks are incomplete



What's New in HRM R9

- New Product Menu
 - Quickly and easily move within the HRM product with single key strokes and short cuts
- 30 new DFSMSHsm “canned” reports
 - HRM R9 now reads the active CDS and/or LOG file to produce real-time reports
- DFSMSHsm internal information
 - DFSMSHsm CSA allocation, usage, etc.
- Additional Storage and DFSMS reporting
 - Report on volumes managed by ADDVOL commands and DFSMS owned volumes

What's New in HRM R9

- New Health Reporting
 - Comprehensive reporting on the overall health of your DFSMSHsm and storage environment
- Display the HSM queues
 - Display and Monitor DFSMSHsm requests
 - Perform actions on behalf on DFSMSHsm
- Cancel HSM requests within the product
 - Perform an action against a DFSMSHsm request

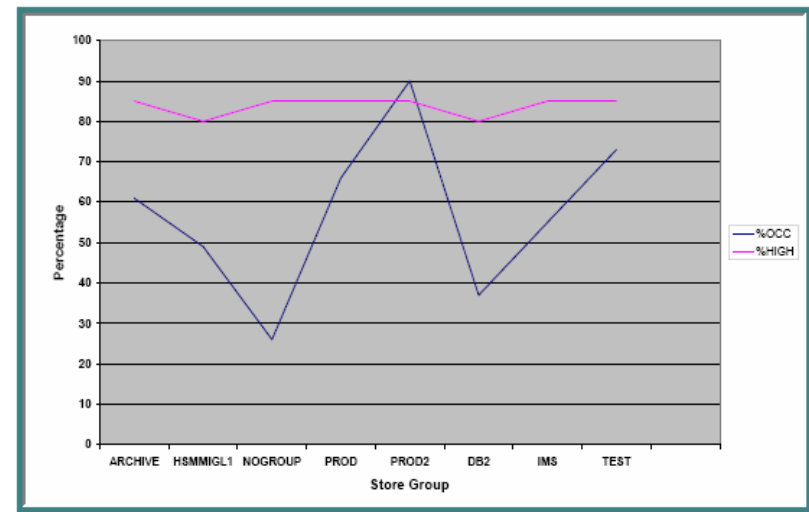
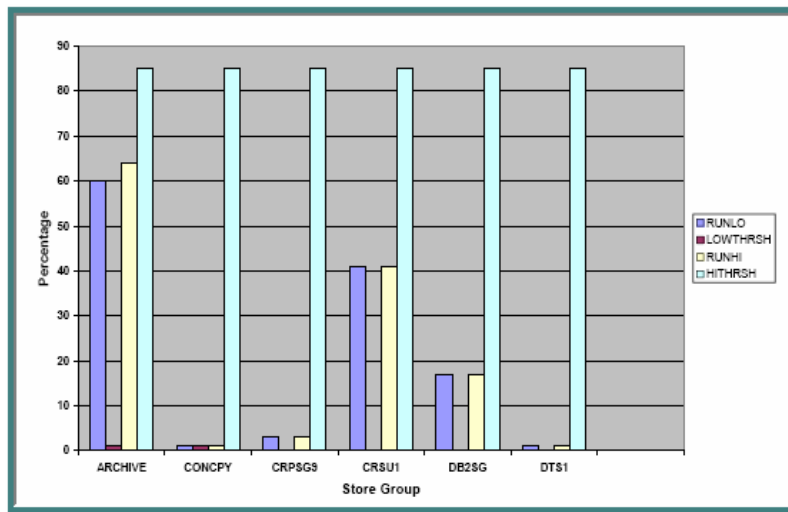
More HRM Features

- ISPF interface
- Tracking and trending reports
 - Predefined Excel graphs
- Format and Save
 - Create custom reports on specific fields
- Filters and Sort
 - Filters can be changed/modified on the fly
 - “Wildcard” filtering capabilities



HRM Features

- Automatic Microsoft Office Excel Charting
 - Advanced Reporting for DFSMShsm utilizes the power of Microsoft Office Excel for charting
 - Charts and graphs are predefined
 - Macros are shipped with the product code
 - There are over 80 charts provided



HRM Features

- Corrective Action
 - Issue Commands to Correct Error Conditions
 - DFSMShsm, IDCAMS, other
 - Select Error Corrections
 - Provided customizable library
 - Create and save your own commands
 - Process Corrective Commands
 - Interactively
 - Batch job

HRM Overview

- Real-time reporting
 - Fast and highly interactive
 - Easily find areas of concern
 - Drill down to view to the area of concern
 - Analyze
 - Act on what you see
- Perform “what-if” analysis
 - Migration thresholds
 - Recycle percent valid



HRM Overview

- DFSMShsm Control Data Set (CDS) Reporting
 - "See" record contents in the CDSs
 - Filter and sort by:
 - Application
 - TSO user
 - Migration status
 - Backup versions
 - Other
 - Customized criteria
 - Identify data sets by creation date
 - Identify by migration date
 - Identify by last used date
 - Identify by size, recall thrashing, etc.



Product Customization

- Product is installed, customized and running in minutes
 - Customize processing parameters
 - Manage batch processing plans
 - Generate JCL for batch processes
 - Test user access to HSM commands
 - Browse member lists for the PARMLIB and plan data sets

```

Menu  Utilities  Navigate  Help
-----
ADMNMENU ----- HSM Reporter/Manager 09.01 ----- 08:58
Command ==> _____ Scroll ==> PAGE

Administrative Menu:                               Row 1 of 12

  Report Name      Description
  -----
1 Customization    Customization Functions
2 Select Plans for Batch  Select Plans to Run in Batch
3 Batch JCL Build   Build JCL for Batch Processing
4 HSM Command Access  Test User Access to HSM Commands
5 List Parmlib Members  Interact with Parmlib Members
6 List PLAN Members   Interact with PLAN Members
    
```

HRM R9 Main Menu



```

Menu Utilities Navigate Help
-----
MAINMENU ----- HSM Reporter/Manager 09.01 ----- 09:32
Command ==> _____ Scroll ==> PAGE

Main Menu: Row 1 of 10

Report Name      Description
-----
1 HSM Host Information      Information on DFSMSHsm Hosts
2 Reports From the CDSs     Reports from DFSMSHsm Control Data Sets
3 Storage Reports           Reports on System Storage
4 DFSMS Information         Information about DFSMS
5 Classic Reports           Classic Reporting Data
6 Administrative Tasks       Administrative Functions
7 Product Information       Product Information
8 Volume Level Data          Volume Level Information
9 Data Set Level Data        Data Set Level Information
    
```

HSM Host Information

- Active HSM Hosts
 - Active CDS and ARCLOG information
 - Data sets allocated to HSM hosts
 - Volumes used in HSM processing
 - Current HSM Work Elements
 - Current SETSYS Processing Values
- HSM Exits in Use
 - Enqueues Held by HSM Hosts
 - Virtual Memory Usage Report
 - Patches Available to DFSMSHsm
 - Automatic Function Data
 - Backup
 - Migration
 - Dump

Storage Reports

- Detail storage information
 - Units available to a system
 - Volume information
 - Storage Group information

```

Menu  Utilities  Navigate  Help
-----
STORMENU ----- HSM Reporter/Manager 09.01 ----- 08:55
Command ==> _____ Scroll ==> PAGE

Storage Reports:                                     Row 1 of 6

  Report Name      Description
  -----
1 Unit Names      Esoteric and Generic Unit Names
2 Volume Information  Volume Level Information
3 Storage Groups   Lists Storage Groups
    
```

DFSMS Reports

- Detail reports on SMS environment

```

Menu Utilities Navigate Help
-----
SMSMENU ----- HSM Reporter/Manager 09.01 ----- 08:35
Command ==> _____ Scroll ==> PAGE

DFSMS Reports: Row 1 of 14

  Report Name      Description
  -----
1 SMS Base Configuration  Shows basic SMS Information
2 Storage Groups         Lists Storage Groups
3 Storage Classes        Lists Storage Classes
4 Management Classes     Lists Management Classes
5 Data Classes           Lists Data Classes
6 Aggregate Groups       Lists Aggregate Groups
7 SMS Volumes            Lists Volumes
    
```

Classic HRM Interface

- Provided by previous versions of HRM

```

Menu  Utilities  Navigate  Help
-----
Classic  ----- HSM Reporter/Manager 09.01 ----- 08:48
Command ==> _____

Support Activities
-----
D. Display Last Master
F. Program Function Keys
J. System JCL Parameters
K. System Data Sets
O. Generate Master

Reports from Control Data Sets
-----
C1. MCDS Information
C2. BCDS Information
C3. OCDS Information
C4. HSM Health Reports
C5. Volume Dump Information

Reports from Log File
-----
1. Daily Space Manager
2. Migration/Backup Options
3. Recall/Recover Options
4. Extent Reduction Summary
5. Recycle Return Code Summary
6. Return Code Reason Code Tables
7. Log Message Summary
8. Tape Usage Summary

Online Batch Process
-----
P. select online Plans for Batch  B. Batch JCL Build for onlines
    
```

Product Information

- Data sets used by HRM during processing
- Data sets allocated to the HRM address space
- Enqueues being held by the product address space
- Virtual storage being used by the product address space

```

PRDMENU ----- HSM Reporter/Manager 09.01 ----- 09:08
Command ==> _____ Scroll ==> PAGE

Product Menu:                                     Row 1 of 10

  Report Name      Description
  -----
1 Version/Release Data    Product Name and Release Data
2 Product Data Sets       Data Sets Used by the Product
3 Address Space Allocations  A/S Active Allocation List
4 Address Space Enq Data   Enqs Held by the Address Space
5 Address Space Memory     A/S Current Memory Usage
    
```

Volume Level Data

- Display and manipulate backup/migration volume information
 - Version information for volumes with full volume dumps
 - Tape volume information
 - Free space information for migration level 1 volumes
 - Primary DASD volume information

```

Menu  Utilities  Navigate  Help
VOLCMD  ----- HSM Reporter/Manager 09.01 ----- 09:14
Command ==> _____ scroll ==> PAGE

Volume Level Reports and Commands:                               Row 1 of 16

  Report/Command      Description
  -----
  DFSMShsm Information:
1 Migration Volumes    View Migration Volumes
2 Backup Volumes       View Backup Volumes
3 Dump Volumes         View Dump Volumes
4 Dump Versions        View Dump Volume Versions
5 Tape Volumes         View Tape Volumes
6 Tape Copy Needed     View Tapes Needing Copying
7 Mounted Volumes     Volumes Used by DFSMShsm
8 ML1 Volume Free Space Migration Level 1 Volume Space

  Other Information:
9 Primary DASD Volumes Volumes on Primary DASD
    
```

Data Level Data

- Display and manipulate information for data sets
 - Migrated, Backed Up, On Tape
 - Migrate, Backup, Recall, Recover, Delete

```

DSNCMD ----- HSM Reporter/Manager 09.01 ----- 09:19
Command ==> _____ Scroll ==> PAGE

Data Set Level Reports and Commands:                               Row 1 of 26

  Report/Command      Description
  -----
  DFSMSHsm Information:

1 Migration Information      View Migration Data
2 Backup Information         View Backup Data
3 Backup Versions           View Backup Versions
4 Tape Information          View Tape Information
5 Dump Information          View Dump Information

  DFSMSHsm Commands:

6 Migrate Data Set          Migrate a Data Set
7 Backup Data Set           Create a Backup Copy
8 Recall Data Set           Recall a Migrated Data Set
9 Recover Backup Copy       Recover a Backed Up Data Set
10 Delete Migration Copy    Delete a Migrated Data Set
    
```

Other Data Set Level Commands

- List Catalog Information
- Delete, browse, view or edit contents of PDS or PS

11	Delete Backup Copy	Delete a Backup Version
Other Commands:		
12	List Catalog Data	Perform a LISTCAT on a Data Set
13	Delete Data Set	Delete a Data Set on Dasd
14	Browse Contents	Browse a QSAM Data Set
15	View Contents	View a QSAM Data Set
16	Edit Contents	Edit a QSAM Data Set



Logfile Summary Reports

MIGRATE ----- HSM Reporter/Manager 09.01 ----- Row 1 to 8 of 8
 Command ==> _____ scroll ==> PAGE

For ALLVOL

S	Rc	Count	Message
	00	39	MIGRATION OK
	06	410	DUPLICATE DSN IN MCDS
	20	30	NOT ELIGIBLE FOR MIGRATION
	30	190	DATA SET NOT CATALOGED
	58	212	MIGRATION OR DBA DBU FAILED
	70	190	SMS DS ERROR
	73	22286	PDSE DSN COULDNT MIGRATE
	99	388	UNSUPPORTED DS

***** Bottom of data *****

Logfile Summary Reports

- Zoom to Detail

```

OPTION ==>
ENTER 0 AT OPTION FOR LIST OF OPTIONS.
PANEL 1 OF 3. SCROLL RIGHT FOR MORE INFORMATION.
KEY S BESIDE ENTRY FOR RETURN/REASON CODES

```

S	DSN	ACTION	RC	RSNC	TRKS	AGE
	BETA93.V4R1M0.LOAD	MIGRATE	99	00014	0	0
	SAI1.CAI.TLOADLIB	MIGRATE	99	00006	0	15
	SYS3.EREP.PPLEX.G4113V00	MIGRATE	99	00004	0	0
	SYS3.PROD3.WLM.PERFINDX.G1332V00	MIGRATE	99	00004	0	675
	SYSG.CSF.SCSFCKDS.V01.DATA	MIGRATE	99	00020	0	0
	SAC1.DENEM	MIGRATE	99	00004	0	211
	STT1.OUT	MIGRATE	99	00004	0	974
	NGK1.SNIF5.SRC	MIGRATE	99	00004	0	588
	CRPROD.CSF.SCSFPKDS.V01.DATA	MIGRATE	99	00020	0	23
	SYSG.TSSAUDIT.D090799	MIGRATE	99	00004	0	15
	SAC1.\$S.LOAD	MIGRATE	99	00014	0	0
	UDS1.USER.SRC	MIGRATE	99	00004	0	1056
	SYS3.SMF.PRDB.TYPE42.G0531V00	MIGRATE	99	00004	0	40
	SYS3.SMF.PRDC.TYPE42.G0353V00	MIGRATE	99	00004	0	0
	SYS3.PROD2.WLM.PERFINDX.G1332V00	MIGRATE	99	00004	0	675
	SYS3.PROD3.WLM.PERFINDX.G1329V00	MIGRATE	99	00004	0	676
	SYS3.PROD2.WLM.PERFINDX.G1327V00	MIGRATE	99	00004	0	677
	SYS3.PROD3.WLM.PERFINDX.G1327V00	MIGRATE	99	00004	0	677

Logfile Summary Reports

- Select Data Set for more detail

```
Refer to DFHSM MESSAGE           for more information
      RETURN CODE
REASONS=> 00004  DSN IS NOT PS, PDS, DA OR VSAM
```

HSM Volume Health Report

- Reports on management functions for DASD volumes
 - Migrate, back up, dump, etc.
 - And what failed

```

Health ----- HSM Reporter/Manager 09.01 ----- Row 1 to 9 of 9
Command ==> _____ scroll ==> PAGE

Enter V to display all detail about volume or B to browse error related detail
S Number Count Message
VOLR000 99 PRIMARY VOLUME IS SMS AND HSM MANAGED.
VOLR001 88 PRIMARY VOLUME IS SMS MANAGED, NOT HSM MANAGED.
VOLR002 7 PRIMARY VOLUME IS HSM MANAGED, NOT SMS MANAGED.
VOLR003 184 MOUNTED VOLUME IS NOT SMS MANAGED, AND NOT HSM MANAGED.
VOLR004 11 MOUNTED VOLUME IS MIGRATION LEVEL 1 VOLUME.
VOLR005 4 MOUNTED VOLUME IS MIGRATION LEVEL 2 VOLUME.
VOLR006 0 MOUNTED VOLUME IS A BACK-UP VOLUME.
VOLR007 4 VOLUME HAS NO ACTIVE VTOC INDEX
VOLR008 0 CACHED VOLUME HAS NO FEATURES ENABLED.
***** Bottom of data *****
    
```

HSM Dump Volume Information Report

- HSM Dump Volume Inventory
 - View dump classes
 - Tape volumes associated with dump
 - DSN info in the VTOC if it exists

```

TapeDSNs ----- HSM Reporter/Manager 09.01 ----- Row 1 to 2 of 2
Command ==> _____ scroll ==> PAGE

      VIEW OF VOLUME DUMPS
Panel 1 of 3.  scroll right(PF11) for VTOC and Tape dsnames
Enter 0 in Command field for options
Select C for dump class detail, D for dsn list, V for list of tapes
   Jul   Greg      Dump  Dump      Tape  CMD or  VTOC CPY
S Date   Date      Time  Volume Class  EXPDT EXPDT  AUTO  Exists
 2008073 03/13/2008 22:11:44 NSMS00 DAILY   2008104 1999000 AUTO   YES
 2008073 03/13/2008 22:11:42 SCD001 DAILY   2008104 1999000 AUTO   YES
***** Bottom of data *****
    
```



CDS Reports

- Realtime – no extractions needed for CDSs
 - Detail reports on backup, migration, etc

```

CDSMENU ----- HSM Reporter/Manager 09.01 ----- 12:25
Command ==> _____ Scroll ==> PAGE

DFSMShsm Reports From Control Data Sets:                               Row 1 of 22

  Report Name      Description
  -----
1 HSM Control Data Sets  shows the HSM Control Data Sets
2 HSM CDS Users         Shows Users of the CDSS

3 Migrated Data Sets    Data Set Level Migration Info
4 Backup Data Sets      Data Set Level Backup Info
5 Backup Versions       Lists Versions of a Backup Ds
6 Tape Data Sets        Lists Data Sets on Tape Volumes
7 Dump Data Sets        Data Sets on Dump Volumes

8 Migration Volumes     Volumes Used in Migrate Commands
9 Backup Volumes        Volumes Containing Backup Data
10 Dump Volumes          These Contain Volume Dump Copies
11 Dump Versions        Lists Versions of Dump Copies
12 Tape Volumes          Tape Volume Information
13 Tape Copy Needed     Tape Volumes Needing Updating
14 ML1 Volume Free Space Free Space for ML1 Volumes

15 Authorized Users     Users with Special HSM Authority
16 Dump Classes          Dump Classes and Attributes
17 Daily Statistics      Daily Statistics Kept by HSM
18 Daily Function Stats  Daily Statistics For Functions
    
```



HRM CDS Reporting

- Migration (MCDS) or Back Up (BCDS) Details
 - By SMS Type and Classes
 - By Migration or Back Up Volume

Menu Utilities Help

MCDS

----- Advanced Reporting for DFSMSHsm V1R1

Option==> _____

Row 1 to 14 of 448

----- 14:10

Scroll ==> PAGE

Enter 0 in option field for options

Panel 3 of 4. Scroll left or right for more information.

C	Dsname	Mig Times	T H L	M L V	N	ML0 Trks	Migrated KB2	Bytes
	N037.CTR.TORDSDT5	19	9	2	N	4050		96845
	N037.CTR.TORDSDT.LD5	17	9	2	N	2175		59126
	N037.CTR.TORDSDT.FORMAT.SORT8	21	9	2	N	1275		34506
	N037.CTR.TORDSDT.ERROR8	21	9	2	N	15		1
	N037.SDAFLOW.SEQ	5	7	2	N	2985		81493
	N037.SDAFLOW.LINK.FORMAT	5	7	2	N	3075		83638
	N037.NDAFLOW.LINK.SORT1	5	7	2	N	930		25029
	N037.GRDFLOW.LINK.SORT2	5	7	2	N	1065		28934
	N037.CTR.TORDSDT.FORMAT6	2	6	2	N	1710		46555
	N037.CTR.TORDSDT.FORMAT4	2	6	2	N	1875		51114
	N052.BAS.N0520M04.OPLD.EXPRD.G1811V00	2	6	2	N	15		1
	N052.BAS.N0520M04.OPLD.EXPRD.G1808V00	2	6	2	N	15		1
	N052.BAS.N0520M04.OPLD.EXPRD.G1799V00	2	6	2	N	15		1
	N052.BAS.N0520M04.DIALS.EXPRD.G1814V00	2	6	2	N	15		1

4B

:00.1

05/12

HRM CDS Reporting

- OCDS Details
 - Portions of the OCDS Tape Inventory or ALL of it

```
Menu  Utilities  Help
-----
Tape Inv ----- Advanced Reporting for DFSMSHsm V1R1 ----- 14:15
Option==> _____

select tape inventory options below

1  Migration Level 2 Tapes
2  Spill Tapes
3  Cycle Tapes (Backup)
4  Unassigned Tapes
5  All Tapes in OCDS Inventory

6  Filter Tape Volume Inventory
7  Display Plans for Tape Volume Inventory
8  Display Filtered Flat Files for Tape Inventory
```

Sort and Filter

- 'O' – Options from most views

```

Menu  Utilities  Help
-----
Options ----- Advanced Reporting for DFSMSHsm V1R1 ----- 14:21
Option==> _____

Enter Option Number
1 Sort
2 Filter
3 Print
4 Save
F Format and Save
5 Plan
6 Total
7 Tag
8 Untag
9 Run
10 Batch
    
```

Sort and Filter

- Data can be sorted on any value listed
 - Ascending or descending
 - Can specify SORT order if multiple fields selected

```

Menu  Utilities  Help
-----
Tape      ----- Advanced Reporting for DFSMSHsm V1R1 ----- 14:27
Option==>
Key N to sort with no duplicates==>
Enter numerical values  A=Ascending  Note: The A or D is required if
For sort order below:  D=Descending  > 1 field is selected.
Vol Type ==> CP=====>
Volser =====> Esoteric=====>
Full =====> Youngest=====>
Dsn(s) =====>
Blks Written=>
Blocks Valid=>
PCT Valid ==> 1 a_
PCT Full ==>
EST CPCTY ==>
Device Type =>
Elig RECYCL =>
Fail RECYCL =>
PRE Vol =====>
SUCC Vol =====>
ALT Vol =====>
Enter REFRESH or R to reset sort to defaults Enter END to cancel sort
Note: It isn't necessary to provide a
sort order number in every field. For
better performance sort only those fields
of interest.

```

Sort and Filter

```

Menu  Utilities  Help
-----
Tape ----- Advanced Reporting for DFSMSHsm V1R1 ----- Row 1 to 13 of 97
Option==> _____ 14:29
                                scroll ==> PAGE

      ARH VIEW OF L2 TAPES
Enter 0 at option for list of options.
Panel 1 of 2. Scroll RIGHT for more information

  Vol
S  Type Volser Full Dsn(s)  Blocks  Blocks  Pct  Pct  EST  Young-
   Type Volser Full Dsn(s)  written valid valid Full CPCTY  est Dsn
L2  210081  N      68    163253      0    000    597    27319  1998171
L2  55555   Y       0         0      0    000    000   1149000  0000000
L2  500010  Y       0         0      0    000    000   1149000  0000000
L2  300051  Y       0         0      0    000    000   1149000  0000000
L2  DAND05  Y        5         6      5    001    000    54638  2001285
L2  210060  N     111     7950    6509    001    000   1149000  1998251
L2  210112  N     117     5652    5215    001    000   1149000  1998251
L2  300038  Y     269     2702     298    001    000   1149000  2006271
L2  300028  N       1         1      1    001    000   1149000  2006277
L2  300010  N       2        548      1    001    000   1149000  2006276
L2  300002  N       2         5      5    001    000   1149000  2006275
L2  210156  Y    1483    924614   46051    004    080   1149000  1998218
L2  210137  Y     678    908279   57416    004    079   1149000  1998220

4B  :00.1 05/12

```

Sort and Filter

- Data can be filtered on any value listed
- Filter operands are:
 - EQ, NE, LT, GT, NH, NL, NI, LE, LK, NK

```

Menu  Utilities  Help
-----
OCDS      ----- Advanced Reporting for DFSMSHsm V1R1 ----- 14:32
Option==> _____

Filter Option==>      Enter F to perform filter in TSO batch
Label      OPCODE      Argument
Vol Type    =>          =>          Device Type =>          =>
Volser      =>          =>          Elig RECYCL =>          =>
Full        =>          =>          Fail RECYCL =>          =>
Dsn(s)      =>          =>          PRE Vol    =>          =>
Blks Written=>          =>          SUCC Vol   =>          =>
Blocks Valid=>          =>          ALT Vol    =>          =>
PCT Valid   => le      => 050      Esoteric   =>          =>
PCT Full    =>          =>          CP          =>          =>
EST CPCTY   =>          =>          Youngest   =>          =>
Enter REFRESH or R to reset filter to defaults
    
```

Sort and Filter



```

Menu  Utilities  Help
-----
Tape ----- Advanced Reporting for DFSMSHsm V1R1 ----- Row 41 to 52 of 52
Option==> _ ----- 14:40
                                           Scroll ==> PAGE

      INTERACTIVE FILTERED VIEW OF OCDS TAPES
Enter 0 at option for list of options.
Panel 1 of 2. Scroll RIGHT for more information

  Vol
S  Type Volser Full Dsn(s)  Blocks  Blocks  Pct  Pct  EST  Young-
   Type Volser Full Dsn(s)  Written  Valid   Valid Full CPCTY  est Dsn
   L2  210103  Y    1501   954678  441781  038   083  1149000 1998243
   L2  210122  Y    1304   791533  449099  039   068  1149000 1998196
   L2  210111  Y    1093   841195  448972  039   073  1149000 1998243
   L2  210142  Y     614   783208  449372  039   068  1149000 1998202
   L2  210031  Y    1094   767330  308195  040   100   767330 1998238
   L2  210131  Y     549   588463  474961  041   051  1149000 1998208
   L2  210130  Y    1371   800055  480010  041   069  1149000 1998195
   L2  210029  Y    1348   955466  403113  042   100   955466 1998238
   L2  210067  Y     465   741260  497134  043   064  1149000 1998166
   L2  210061  Y     957   701664  530208  046   061  1149000 1998131
   L2  210014  Y    1631   806615  398186  049   952   27319 1998139
   L2  210106  Y     387   733152  569684  049   063  1149000 1998119
***** Bottom of data *****

```

Corrective Actions

- Example: Migrate Data Sets Now
 - Enter Option 9 to RUN Command to Migrate

```
Menu Utilities Help
-----
Options ----- Advanced Reporting for DFSMSHsm V1R1 ----- 14:21
Option==> _____

Enter Option Number
1 Sort
2 Filter
3 Print
4 Save
F Format and Save
5 Plan
6 Total
7 Tag
8 Untag
9 Run
10 Batch
```

Corrective Actions

- Advanced Reporting for DFSMSHsm Line Command Processor Panel
 - Press PF05 for a list of Labels for Filter Criteria

```

Menu  utilities  Help
-----
LineCmds ----- Advanced Reporting for DFSMSHsm V1R1 ----- 15
Option==> _____

                                Line Command Processor
Hit PF04 For format options          Hit PF05 To display labels
Hit PF06 To retrieve a list of commands, Hit PF07 To save this command
Title==> MCDS VIEW BY VOLUME B61004
Format options follow
      FORMAT OPTIONS COMP NOSTRING NODICT TSO NOTOTALS NOTITLES
      /ACTION /DSN
    
```

Corrective Actions

- Labels are displayed

```

Menu Utilities Help
-----
Tags vls ----- Advanced Reporting for DFSMSHsm V1R1 ----- 15
Option==> _____

Enter the positional tag you wish to substitute below
Tag==>
SDSN      DATA SET NAME      SMCLSS    MANAGEMENT CLASS
SMG       MIGRATED Y,R,D,P     SSCLSS    STORAGE CLASS
SRCNT     RECALL COUNTS        SDP       SDSP INDICATOR
SCRDT     CREATION DATE       SI        THRASHING INDEX
SRFDT     LAST REF DATE
SMGDT     MIGRATION DATE
SBKDT     BACK-UP DATE
SXPDT     EXPIRATION DATE
SVOL      MIGRATION VOLUME
SL0       PRIMARY VOLUME
SL        MIGRATION LEVEL
SV        VSAM OR NON VSAM
SKB2      KB2 BYTES MIGRATED
STRK      PRIMARY TRACKS
    
```

Corrective Actions

- Advanced Reporting for DFSMSHsm Line Command Processor Panel
 - Key HMIGRATE '/SDSN' in the command line
 - Press 'Enter' and command is sent to HSM

```

Menu  Utilities  Help
-----
LineCmds ----- Advanced Reporting for DFSMSHsm V1R1 ----- 15
Option==> _____

                                Line Command Processor
Hit PF04 For format options      Hit PF05 To display labels
Hit PF06 To retrieve a list of commands, Hit PF07 To save this command
Title==> MCDS VIEW BY VOLUME B61004
Format options follow
      FORMAT OPTIONS COMP NOSTRING NODICT TSO NOTOTALS NOTITLES
      HMIGRATE '/SDSN' _
    
```

Corrective Actions

- Advanced Reporting for DFSMShsm Line Command Processor Panel
 - HSM responds back to your terminal
 - Press 'Enter' when three asterisks are displayed

```
ARC1007I MIGRATE REQUEST 00001257 SENT TO HSM  
***
```

PLANS

- Pre-defined reports provided called PLANS
 - Save your SORT and FILTER criteria
 - Convenient method of saving and executing reports
 - Can invoke extended filtering capabilities
 - Example is to filter on DSNNAME and ACS value
 - View most current data
 - Plans are associated with data items
 - Stored Plans can be
 - Deleted, Edited, Browsed
 - Run interactively or in batch
 - Easily create and modify plans

Plans Member List

```

Menu  utilities  Help
-----
Plans  ----- Advanced Reporting for DFSMSHsm V1R1 ----- Row 1 to 12 of 14
Option==> _____ 14:41
                                scroll ==> PAGE

Specify Sort Key==>          MLIST, TITLE, CREDIT or UPDT
Specify Sort Type==>        A for Ascending or D for Descending
You may Browse (B), Edit (E), View (V), Print (P), Flat (F), Sum (IF MSGS)
or Delete (D)

Command  Member  Action  Title
MCDS@LBL  MCDS@LBL  FIELDS FOR MCDS PLANS
MCDS@LBL  MCDS@LBL  SPECIAL PLAN TO PRINT THE SMS SUMMARY INFO
MCDS@LBL  MCDS@LBL  DISPLAY MIGRATED DATA SETS 2 YEARS OLD
MCDS@LBL  MCDS@LBL  ARH102 CURRENT WEEK MIGRATION THRASHING REPORT
MCDS@LBL  MCDS@LBL  LIST ALL DATA SETS REFERENCED LAST YEAR
MCDS@LBL  MCDS@LBL  DISPLAY THRASHING DSNAMES FOR LAST SEVEN DAYS
MCDS@LBL  MCDS@LBL  DISPLAY DATA SETS MIGRATED AND RECALLED THIS WEEK
MCDS@LBL  MCDS@LBL  DISPLAY DSN(S)W/RECALL COUNT GT 30
MCDS@LBL  MCDS@LBL  PRINT MIGRATION THRASHING FOR LAST 14 DAYS
MCDS@LBL  MCDS@LBL  PRINT DATA SETS TO EXPIRE IN 100 DAYS
MCDS@LBL  MCDS@LBL  SC - TEST PLAN
MCDS@LBL  MCDS@LBL  FORMAT / SAVE WHO RECALLED DATA - SC - REVIEW
    
```

Plan Criteria

```
EDIT      ARH0101X.QABETA.PLANS(MCDS011) - 01.02      Columns 00001 00072
Command ==> _____ Scroll ==> PAGE
***** ***** Top of Data *****
000001 DISPLAY DATA SETS MIGRATED AND RECALLED THIS WEEK
000002 SORT FOLLOWS NODUPS
000003 SMGDT,C,A,SDSN,C,A,STRK,C,A
000004 FILTER FOLLOWS
000005 ASMGDT   NL *-7                                AND
000006 ASMG     EQ R                                AND
***** ***** Bottom of Data *****
```



HRM Benefits

- Gathers Critical Information from DFSMSHsm
- Centralizes and Simplifies DFSMSHsm Management
- Real-time Reporting
- Interactive Display
- Provides Daily and Historical Reporting
- Powerful Filtering and Masking Support for Reporting
- Proactive Analysis Tools
- Ability to Identify and Correct Error Conditions
- Health Reports



Why You Need FastAudit



maxim
ove agility save time
enges colleagues



Auditing the DFSMSHsm Environment

- Reasons for Auditing Control Data Sets
 - DFSMSHsm CDS recovery
 - Periodic audits reduce error conditions in recovered CDS
 - Recovery of an ML1 volume
 - Loss of an ML2 or backup tape
 - Tapes that fail recycle
 - Inability to recall or recover data
 - DFSMSHsm ABENDs or a system crash
 - Recovery of an SDSP

Auditing the DFSMSHsm Environment

- **More** Reasons for Auditing Control Data Sets
 - Aggressive migration policies
 - Reliance on migrated or backed-up data at home and at the disaster recovery site
 - Use of migrated data in production batch
 - Production output files used as input to subsequent processing cycles
 - *Weekly, monthly, quarterly, annual, etc.*
 - Use of ABARS for application backup and recovery

Auditing the DFSMShsm Environment

- And Even **More** Reasons for Auditing
 - Reclaiming tape
 - Using migration as a long-term data store
 - Historical data that is critical to the business
 - *Drawings, legal records, medical information*
 - Regulatory & legal requirements



Why you need *FastAudit*

- Ensure access to DFSMShsm managed data
 - Data can be recalled from migration
 - Data can be restored from DFSMShsm backup
 - Migration and backup tapes can be recycled
 - Corrupted DFSMShsm tape records can be rebuilt
 - DFSMShsm and the tape management system are synchronized
 - Control data sets are healthy and free of errors
 - Data can be recovered at the disaster recovery site



FastAudit Overview

- All audits are comprehensive and *fast*
 - Eliminates the need to execute multiple audit commands for a single control data set
- Automated user-selected fixes
 - Select fixes from a provided list or create your own
 - Errors and fixes are organized by error type in an ISPF panel
- Fix JCL and commands are automatically generated
 - Apply fixes to one, some, or all data sets in a problem class

FastAudit Overview

- Audits for the MCDS, BCDS and OCDS
 - Verifies contents
 - Identifies inconsistencies and errors
 - Compares contents to catalogs and tape management inventory
- Quickly verifies the integrity of DFSMShsm metadata
 - The ability to recall data
 - The ability to recover data
 - The ability to recycle tapes
 - Identifies out-of-sync conditions with other metadata

FastAudit Primary Option Panel

```
<V09 R01> FAST AUDIT.  CHOOSE WHICH AUDIT TO FIX
COMMAND==>
```

- 1 MCDS CREATED BY PROCS FAUDITMD, FAUDITCT OR FAUDITAL
- 2 BCDS CREATED BY PROC FAUDITBD
- 3 OCDS CREATED BY PROC FAUDITOD
- 4 TAPE CREATED BY PROCS FAUDITTP, FAUDITZR, FAUDITRM OR FAUDITHL
- 5 CAT CREATED BY PROC FAUDITCO
- 6 DASD CREATED BY PROC FAUDITDA OR FAUDITDE
- 7 MASTCAT CREATED BY PROC FAUDITMC
- 8 CATRMM CREATED BY PROC FAUDITCR
- 9 CATTMS CREATED BY PROC FAUDITTC
- 9A CATTLMS CREATED BY PROC FAUDITLC
- 10 MEDCTL CREATED BY PROC FAUDITFD, FAUDITFL OR FAUDITFT

FastAudit Error Summary Panel

- Error Suffix Identifies Type of Error
 - **C – Critical**
 - W – Warning
 - V – Volume
 - I - Information

```
<V09 R01> FASTAUDIT MCDS SUMMARY
OPTION ===>
ENTER F FOR FIXS, B BROWSE ERRORS, E EDIT, M MODIFY, H HELP
S NUMBER      COUNT      MESSAGE
11D1C          0      MCD ENTRY IS NOT CATALOGED
11V1C          0      MCO VSAM COMPONENT IS NOT CATALOGED
11D2C          2      MCD ENTRY IS CATALOGED ON DIFFERENT VOLUME
11V2C          2      MCO VSAM COMPONENT ON DIFFERENT VOLUME
1103W          0      MCD ENTRY IS MISSING THE MCA ENTRY
1104C          0      MCD LEVEL 1 ENTRY HAS NO VTOC ENTRY ON L1 VOLUME
1105W          2      MCD VSAM BASE NAME IS MISSING IN MCO
1106C          0      MCD IS ON VOLUME WHICH HAS NO MCV ENTRY
1106V          0      SUMMARY OF VOLUMES HAVING NO MCV ENTRY
1107W          2372    MCD LEVEL 2 IS ON VOLUME WHICH HAS NO TTOC ENTRY
1107V          1      SUMMARY OF VOLUMES MISSING IN TTOC
1108W          0      MCA ENTRY IS MISSING MCO VSAM COMPONENT ENTRY
1109W          0      MCD ALIAS IS NOT ON TTOC
1109V          0      SUMMARY OF VOLUMES WHERE ALIAS IS NOT IN TTOC
1110W          0      MCD HAS DUPLICATE MCA ENTRY
1111W          1      MCA ENTRY IS MISSING MCD ENTRY
1112W          0      MCO VSAM BASE NAME IS MISSING IN MCD
1113W          0      MCO VSAM COMPONENT NAME IS MISSING IN MCA
1114W          2      ALIAS NAME EXISTS ON VTOC, MCA IS MISSING
```

Row 1 to 19 of 31

FastAudit Data Set Error Panel

- Browse list of data sets in error
 - List can be edited

```

Menu  Utilities  Compilers  Help

BROWSE      SK01.H091619                      Line 00000000 col 001 080
Command ==> _____ Scroll ==> CSR
***** Top of Data *****
11D1C CRTSMH.CAT114.MIG.KSDS1                IN-A-SDSP-ON-VOLUME ML1000
11D1C EDP.LAB.MIG2                          HSM.HMIG.T392609.EDP.LAB.J320
11D1C DANR.MQL.OUTPUT                       IN-A-SDSP-ON-VOLUME ML1000
11D1C MH.GTF.TRACE.G0197V00                 HSM.HMIG.U151200.MH.GTF.J7327
11D1C DANRDO11.CRTS01.VSAM                  IN-A-SDSP-ON-VOLUME ML1000
11D1C MH.GTF.TRACE.G0199V00                 HSM.HMIG.T461200.MH.GTF.J7326
11D1C MH.GTF.TRACE.G0200V00                 HSM.HMIG.T371200.MH.GTF.J7326
11D1C MH.SRC#P2S.ASM                       HSM.HMIG.T441200.MH.SRC#P2S.J
11D1C DANR.SI52.NOMCDX                     IN-A-SDSP-ON-VOLUME ML1000
11D1C CRTSMH.CAT114.MIG.KSDS3               IN-A-SDSP-ON-VOLUME ML1000
11D1C CRTSMH.CAT114.MIG.NVSM1               IN-A-SDSP-ON-VOLUME ML1000
11D1C CRTSMH.CAT114.MIG.NVSM2               IN-A-SDSP-ON-VOLUME ML1000
11D1C C21272-1.VSAM01                      IN-A-SDSP-ON-VOLUME ML1000
11D1C DCD.TEST.GDG.G0018V00                 HSM.HMIG.T001400.DCD.TEST.J21
11D1C DANRDO11.TOB.E.ARCIVE                 IN-A-SDSP-ON-VOLUME ML1000
11D1C DCD.TEST.GDG.G0069V00                 HSM.HMIG.T001400.DCD.TEST.J21
11D1C DCD.TEST.GDG.G0142V00                 HSM.HMIG.T001400.DCD.TEST.J21
11D1C CRTSMH.CAT114.MIG.KSDS2               IN-A-SDSP-ON-VOLUME ML1000
11D1C CRTSHSM.VSAM.DATASET                 IN-A-SDSP-ON-VOLUME ML1000

[4B]                                     :00.1                                04/15
    
```

FastAudit Fix Error Panel

- One or more fixes available for each error

```

<V09 R01>
FAST AUDIT FIX PLAN MEMBER LIST
Row 1 to 6 of 6
OPTION ==> SCROLL==> PAGE
YOU MAY BROWSE (B), EDIT (E), FIX (F) HELP (H) OR VIEW (V) RESULTS
COMMAND MEMBER ACTION TITLE
F11D1CA TSO CATALOG A MIGRATED DATASET TO VOLSER MIGRAT
F11D1CB TSO HSEND DELETE MIGRATED DSN MISSING CATALOG
F11D1CC TSO LISTC FOR DATA SET NOT CATALOGED
F11D1CD HSA DISPLAY DUMP OF MCD RECORD
F11D1CR1 HSM DELETE MCD RECORD WHEN UNABLE TO CATALOG MIGRAT
F11D1CR3 HSM FIXCDS PATCH ASSIGNED AND NEEDS SCRATCH
***** Bottom of data *****
    
```

FastAudit Fix Generation

- Fix JCL is automatically generated

```

File Edit Edit_Settings Menu Utilities Compilers Test Help
EDIT SK01.H091228 columns 00001 00072
Command ==> Scroll ==> CSR
000012 //STEP1 EXEC PGM=IKJEFT01,
000013 // DYNAMNBR=25,REGION=4096K
000014 //*
000015 //SYSPROC DD DSN=AKD0201X.QABETA.SAKDCLST,DISP=SHR
000016 //* ****
000017 //SYSOUT DD SYSOUT=*
000018 //SYSTSPRT DD DSN=SK01.FIXAUDIT.F11D1CA,
000019 // DISP=(,CATLG),UNIT=SYSDA,
000020 // SPACE=(CYL,(10,20)),
000021 // DCB=(RECFM=FB,LRECL=133,BLKSIZE=13300)
000022 //SYSTSIN DD *
000023 DEF NVSAM +
000024 (NAME('CRTSMH.CAT114.MIG.KSDS1') +
000025 VOL(MIGRAT) DEVT(SYSALLDA) +
000026 RECATALOG)
000027 DEF NVSAM +
000028 (NAME('EDP.LAB.MIG2') +
000029 VOL(MIGRAT) DEVT(SYSALLDA) +
000030 RECATALOG)
000031 DEF NVSAM +

```

4B :00.1 07/02

Auditing Tape

- AUDITMEDIACONTROLS Audits Tape
- During DFSMShsm Migration TTOC record changes are accumulated in storage
- If you cancel DFSMShsm, or it is otherwise halted, migration tape is put in failed create status and cannot be recycled
- Identify tapes in Failed Create or Failed Recycle status using the LIST TTOC command
- Audit Mediacontrols consumes resources

Auditing Tape

- DFSMShsm Resources
 - Additional resources are required when executing AUDITMEDIACONTROLS
 - Time
 - Several hours, possibly several days, to audit even one tape
 - DFSMShsm
 - CDS backup is held until tape audit is complete
 - Audit executes as a low priority task
 - Hardware
 - Tape drives held and unusable during audit

DFSMSHsm AUDITMEDIACONTROLS

- Can take hours, even days, to audit just one tape

Insurance	3 days for one 3590 MagStar
Trucking	1+ days for one 9840 STK
Bank	1-2 days per 3590 (256 track)
Utility	1-3 days for 3490E tape
Vendor	1-2 days for 3590
Distributor	0% valid data – 7 hours to days
Investment	Up to and exceeding 5 days per tape

SLOOOOOWW...



FastAudit MediaControls

- Overview

- Audit tape contents quickly
- Identifies tape corruption before it is too late
- No significant increase in system usage
- No impact to DFSMShsm request queues
- No hardware tied up for extended periods of time
- Select and prioritize corrective actions

FastAudit MediaControls

- Audit Tape Management Inventory
 - Compares tape management contents to catalog environment to insure correctness
 - Finds active DFSMSHsm tapes that are not cataloged
 - Finds file and tape sequence errors
 - Automatically builds corrective action
 - *Commands*
 - *Batch jobs*



FastAudit MediaControls Error Summary

Messages

AKD720 ----- Row 1 to 16 of 16
 Option==> _____ scroll ==> CSR

You may Browse (B) Edit (E) Fix (F) Help (H) or Modify (M) Results

S Number	Count	Message
8000I	0	LIST OF DATA SETS RECORDED IN MCDS/BCDS
8001C	0	NO HSM REFERENCE FOUND FOR REQUESTED TAPE
8C02C	0	NO MCT ENTRY WAS FOUND FOR REQUESTED TAPE
8D02C	0	NO MCV ENTRY WAS FOUND FOR REQUESTED TAPE
8003C	0	NO TTOC EXTENSION 0000 WAS FOUND FOR VOLSER
8004W	0	VOLUME IS IN FAILED CREATE STATUS
8005W	1	VOLUME IS IN FAILED RECYCLE STATUS
8006C	0	TTOC EXTENSION RECORDS ARE OUT OF SEQUENCE
8007C	0	TTOC EXTENSION RECORD IS CONTAMINATED
8008C	0	TTOC EXTENSION RECORDS DO NOT MATCH EXPECTED COUNT
8009W	0	16K BLOCK COUNT DOES NOT MATCH COUNTED BLOCKS
8C10C	0	MCC EXISTS, NOT ON TAPE
8D10C	0	MCD EXISTS, NOT ON TAPE
8C12C	0	MCC MISMATCH FILE BLOCK IDENTIFIER
8D12C	0	MCD MISMATCH FILE BLOCK IDENTIFIER
8020I	302	LIST OF DATA SETS ON TAPE

***** Bottom of data *****

FastAudit MediaControls Fix Error Panel

- One or more fixes available for each error

```

Messages
-----
AKD715 ----- Row 1 to 3 of 3
Option==> _____ scroll ==> CSR

You may Browse (B) Edit (E) Fix (F) Help (H) or View (V) Results
Command Member Action Title
      F8D10CA      TSO HSEND DEL DATA SET FROM MCDS
      F8D10CB      TSO HSEND RECOVER DATASET FROM BACKUP
      F8D10CC      TSO DISPLAY DATASET FOR BACKUP AND MIGRATION INFO
***** Bottom of data *****
    
```

FastAudit MediaControls Fix Generation

- Fix JCL is automatically generated

```

File Edit Edit_Settings Menu Utilities Compilers Test Help
EDIT      SK01.H124355                                columns 00001 00072
Command ==> _____ scroll ==> CSR
000012 //STEP1 EXEC PGM=IKJEFT01,
000013 //  DYNAMNBR=25,REGION=4096K
000014 //*
000015 //SYSPROC DD DSN=AKD0201X.QABETA.SAKDCLST,DISP=SHR
000016 //*          ****
000017 //SYSOUT DD SYSOUT=*
000018 //SYSTSPRT DD SYSOUT=*
000019 //SYSTSIN DD *
000020 HSEND FIXCDS D DCD.MIGRAT.DONT.RECALL.G0001V00 +
000021 PATCH(40 X'4000') +
000022 PATCH(64 TS0001) +
000023 ODS('SK01.FIXAUDIT.F8020IA') LOGONLY
000024 HSEND FIXCDS D DCD.MIGRAT.NEVER.RECALL.G0094V00 +
000025 PATCH(40 X'4000') +
000026 PATCH(64 TS0002) +
000027 ODS('SK01.FIXAUDIT.F8020IA') LOGONLY
000028 HSEND FIXCDS D DCD.MIGRAT.NEVER.RECALL.G0095V00 +
000029 PATCH(40 X'4000') +
000030 PATCH(64 TS0001) +
000031 ODS('SK01.FIXAUDIT.F8020IA') LOGONLY
4B  @:00.1 04/15
    
```

FastAudit Benefits

- Audits DFSMShsm managed tapes quickly
- Identifies tape corruption before it is too late
- No significant increase in system usage
- No impact to DFSMShsm request queues
- No hardware tied up for extended periods of time
- Select and prioritize corrective actions

Conclusion

So...It is Important to Audit DFSMShsm Regularly!

- Errors creep into DFSMShsm control data sets more often than you think!







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